

Bayesian Decision Theory

1. How does Bayes' theorem form the foundation of Bayesian classification? Derive the formula and explain each component.

Answer:

Bayes' theorem is the core of Bayesian classification. It provides a systematic way to update the probability estimate for a hypothesis (class) based on observed data (features).

Bayes' Theorem Formula:

$$P(w_j|x) = \frac{p(x|w_j)P(w_j)}{p(x)}$$

Where:

- $P(w_j|x)$: Posterior probability of class w_j given observation x
- $p(x|w_j)$: Likelihood – probability of observing x given class w_j
- $P(w_j)$: Prior probability of class w_j
- $p(x)$: Evidence – total probability of observing x across all classes

Explanation:

- The theorem transforms our prior belief $P(w_j)$ into a posterior belief $P(w_j|x)$ after seeing evidence x .
- The evidence term ensures normalization so that the posterior probabilities sum to 1.

This theorem enables **informed classification** by combining prior expectations and observed data in a probabilistic framework.

2. What is class-conditional probability, and how does it differ from prior and posterior probabilities? Provide mathematical representation and interpretation.

Answer:

Class-conditional probability $P(x|w_j)$ is the likelihood of observing feature x given that the actual class is w_j .

Mathematically:

$p(x|w_j)$ = Probability density of feature x if the class is w_j

Contrast with:

- **Prior $P(w_j)$:** How probable the class is without any feature observation.
- **Posterior $P(w_j|x)$:** How probable the class is *after* observing x .

Interpretation:

It reflects the distribution of data within each class. In fish classification, $p(x|w_1)$ might represent the distribution of lightness values for sea bass.

Use:

It's crucial in computing the posterior via Bayes' theorem and in determining decision boundaries.

3. Compare and contrast minimum-risk classification and minimum-error-rate classification. Include examples where each is appropriate.

Answer:

- **Minimum-error-rate classification** assumes all errors are equally bad (zero-one loss), and chooses the class with the **maximum posterior probability**.
- **Minimum-risk classification** incorporates a **loss function** to weight errors differently and chooses the action with the **lowest expected loss**.

Example:

- Spam filtering with equal costs = minimum-error-rate.
- Medical diagnosis (e.g., false negatives being more costly) = minimum-risk classification.

4. What is conditional risk in Bayesian classification, and how does it influence the decision rule?

Answer:

Conditional risk $R(\alpha_i|x)$ is the expected loss when taking action α_i after observing feature vector x . It's defined as:

$$R(\alpha_i|x) = \sum_{j=1}^c \lambda(\alpha_i|w_j)P(w_j|x)$$

It influences decision-making by guiding us to choose the action (e.g., class) with the **minimum conditional risk**. This ensures that decisions are not just based on which class is most likely, but also on how costly wrong decisions are.

5. Discuss how the ROC curve helps evaluate the performance of a binary Bayesian classifier.

Answer:

The **ROC (Receiver Operating Characteristic) curve** is a plot that shows the trade-off between:

- **True Positive Rate (TPR)** — sensitivity or recall
- **False Positive Rate (FPR)** — 1 - specificity

By adjusting the decision threshold in a Bayesian classifier (e.g., the posterior probability cutoff), different points on the ROC curve can be generated.

Benefits:

- Useful when dealing with **imbalanced datasets**
- Highlights the trade-offs between **catching positives** and **avoiding false alarms**
- The **Area Under the Curve (AUC)** provides a **single metric** for overall performance:
 - AUC = 1: perfect classifier
 - AUC = 0.5: random guessing

6. Define Bayes risk and Bayes error. Why are these considered benchmarks for classification performance?

Answer:

- **Bayes risk** is the **minimum possible risk** (expected loss) achievable by any classifier under known distributions.
- **Bayes error** is the **lowest achievable probability of misclassification**, found by applying the Bayes decision rule with zero-one loss.

These benchmarks define the **theoretical limit** of classification performance, and all real-world classifiers aim to approximate or approach them.

7. Compare linear and quadratic discriminant analysis (LDA vs QDA) from a Bayesian perspective. When is each appropriate?

Answer:

Both LDA and QDA are derived from **Bayesian classification using Gaussian likelihoods**.

Linear Discriminant Analysis (LDA):

- Assumes all classes share the **same covariance matrix**.
- Discriminant functions are **linear** in x .
- Decision boundaries are **hyperplanes**.

Quadratic Discriminant Analysis (QDA):

- Allows each class to have its **own covariance matrix**.
- Discriminant functions are **quadratic** in x .

- Decision boundaries are **curved** (hyperquadrics).

When to Use:

- **LDA**: When the assumption of equal spread is valid; less data needed.
- **QDA**: When classes have significantly different variances or orientations.

Trade-off:

- LDA is simpler and requires fewer parameters.
- QDA is more flexible but prone to overfitting with limited data.

8.How does the choice of covariance matrix affect decision boundaries in Gaussian classifiers?

Answer:

The form of the **covariance matrix** Σ in a Gaussian classifier directly affects the **shape of the decision boundaries**:

- **Case 1:** $\Sigma_i = \sigma^2 I$
 - Covariances are equal and isotropic (spherical).
 - **Linear boundaries** perpendicular to the line between means.
- **Case 2:** $\Sigma_i = \Sigma$
 - Covariances are equal but not isotropic.
 - Still **linear**, and Hyperplane passes through x_0 but is not necessarily orthogonal to the line between the means
- **Case 3:** $\Sigma_i = \text{arbitrary}$
 - Each class has its own covariance matrix.
 - **Quadratic boundaries** – can curve and adapt to complex data.
 - Decision boundaries are hyperquadric

This choice influences **flexibility**, **computational complexity**, and the **risk of overfitting**.

1) What does $p(x|w_j)$ represent in Bayesian Decision Theory?

- A. Posterior probability
- B. Joint probability
- C. Class-conditional probability
- D. Evidence

Answer: C

2) What does Bayesian Decision Theory primarily quantify?

- A. Neural network performance
- B. Tradeoffs between decisions using probabilities and costs
- C. Image recognition accuracy
- D. Reinforcement learning rewards

Answer: B

3) The Bayes decision rule chooses the class with:

- A. The smallest prior probability
- B. The highest loss
- C. The largest posterior probability
- D. The lowest likelihood

Answer: C

4) What is the goal of minimum-risk classification?

- A. Maximize error
- B. Minimize computational complexity
- C. Minimize the expected loss
- D. Maximize posterior probability

Answer: C

5) In two-category classification, a decision is made based on:

- A. Mean of feature values

- B. Prior only
- C. Likelihood ratio compared to a threshold
- D. Variance of data

Answer: C

6) Which distribution has the maximum entropy for given mean and variance?

- A. Binomial
- B. Gaussian
- C. Poisson
- D. Uniform

Answer: B

7) What does the zero-one loss function assume?

- A. All decisions are correct
- B. Errors are unequally penalized
- C. All errors are equally costly
- D. Only one state exists

Answer: C

8) What does the ROC curve show?

- A. Tradeoff between false positive and true positive rates
- B. Classification boundaries
- D. Training error over epochs
- D. Posterior distribution over time

Answer: A

9) What is the Bayes error?

- A. Maximum classification error
- B. Error with maximum loss

- C. Minimum possible error for any classifier
- D. Total misclassified data points

Answer: C

10) The Mahalanobis distance is used in multivariate Gaussian models to measure:

- A. Euclidean similarity
- B. Squared error loss
- C. Distance normalized by variance and correlation
- D. Risk difference between decisions

Answer: C